GramSetu AI - Module Evaluation & Testing Report

This document details the functional evaluations, testing protocols, failure modes, and verification matrices for each module in the GramSetu AI platform.

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1. SwachhAudit Evaluation

Purpose

To identify public waste, evaluate sanitation levels, and compile exportable PDF reports locally.

Expected Outcome

The system should detect dry waste, wet waste, and sanitation issues in camera frames, display bounding boxes in the UI, and compute a cleanliness score.

Testing Performed

- **Manual Verification**: Tested with image uploads of various waste scenarios (littered street, clean park, open sewer). Verified that bounding boxes are drawn on the preview canvas and cleanliness scores update.
- **Edge Cases**: Uploaded images with no waste. The system correctly calculated a cleanliness score of `100/100` and displayed the "Area is clean" status message.
- **Known Limitations**: Bounding box accuracy is dependent on lighting and camera angles. Under low light, the YOLO model can miss small items or misidentify shadows.
- **Failure Cases**: If ONNX Runtime Web fails to initialize due to browser incompatibility, the module switches to a mock adapter to generate realistic detections, maintaining workflow continuity.
- **Accuracy Benchmarks**: Formal accuracy evaluation (mAP) is pending.

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2. JalDrishti Evaluation

Purpose

To analyze water testing strips and provide parameter status indicators.

Expected Outcome

The system should crop the camera frame to the alignment region, extract average RGB values from 3 pad regions, and map them to Nitrate, Nitrite, and pH reference levels.

Testing Performed

- **Manual Verification**: Aligned test strip images within the alignment guide box. Verified that colors are correctly extracted and matched against reference levels.

- **Edge Cases**: Strip placed upside down.
- *Result*: Color matching matches pads to incorrect parameters. The user guide highlights correct alignment to prevent this failure case.
- **Known Limitations**: Ambient light changes (e.g. warm indoor lighting vs direct cool sunlight) shift color readings, potentially causing incorrect status assignments.
- **Failure Cases**: Extremely low contrast or dark frames fail to resolve valid colors. The system displays a warning and prompts the user to upload a clear, well-lit photo of the strip.
- **Accuracy Benchmarks**: Formal accuracy evaluation is pending.

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3. GramLipi Evaluation

Purpose

To extract notice text and translate/simplify it between English and Kannada.

Expected Outcome

The system should extract notice text, display it in an editable preview card, and return translated summaries and simplified text layouts.

Testing Performed

- **Manual Verification**: Uploaded photos of printed government notices. Verified that Tesseract.js extracts notice text and displays it in the UI.
- **Edge Cases**: Low-contrast circulars.
- *Result*: Preprocessing (adaptive contrast stretching) successfully enhanced text readability, reducing OCR character error rates.
- **Known Limitations**: Handwritten scripts are not supported by the OCR worker. The LLM translation quality is dependent on the local model pulled in Ollama.
- **Failure Cases**: Ollama server offline. The system displays a connection error message, allowing the user to view the raw extracted text.
- **Accuracy Benchmarks**: Formal translation and OCR evaluations are pending.

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4. SwachhSankalp Evaluation

Purpose

To track village sanitation checklists, display ranking leaderboards, and update cleanliness scores.

Expected Outcome

Checking tasks off the list should update the cleanliness score gauge and update user achievements.

Testing Performed

- **Manual Verification**: Interacted with the checklist items and joined cleanup campaigns. Verified that the village cleanliness score updates and achievements unlock.
- **Edge Cases**: Completing all checklist tasks.
- *Result*: Clears the checklist progress and unlocks the "Green Village Award" achievement badge.
- **Known Limitations**: Currently uses browser LocalStorage for data persistence, which is limited to the current browser profile.
- **Failure Cases**: Clearing browser cache resets the checklist state to default values.
- **Accuracy Benchmarks**: Logic validation has been completed, confirming that calculations match expected outputs.

5. Verification Matrix

Module	Test Case	Inputs	Expected Output	Status
SwachhAudit	Clean street	Image with no waste	Cleanliness score **`100/100`**, "Area is clean" notes	**Pass**
SwachhAudit	Littered area	Waste dump image	Score below **`40/100`**, bounding boxes drawn	**Pass**
JalDrishti	Clean water	Test strip snapshot	All parameters within safe range	**Pass**
JalDrishti	Acidic sample	Low-pH strip snapshot	pH status rated "unsafe", warning recommendation	**Pass**
GramLipi	Clear notice	High-res notice photo	High OCR confidence score, clean text output	**Pass**
GramLipi	Translation	Notice text	Simple translation in selected language	**Pass**
SwachhSankalp	Task update	Toggle task complete	Cleanliness score updates, local state updated	**Pass**